



Tecnologico de Monterrey
Interpartial Exam 1 A
2026-Jan-May

Grade

Course: Calculus I

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 26/01/2026

Last Names and Names: _____

Group: 201

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Algebra Review. Read the following questions and justify your answer with a procedure.

1. (13 points) Solve the following equation,

$$f(x) = e^{2x} - 3e^x = -2$$

2. (12 points) Find the asymptotes and intersections of the following function,

$$g(x) = \frac{4x^2 - 1}{2x^2 - 1}$$

3. (12 points) Consider that $h(x) = e^x$ and $j(x) = 1 - x$. Find the following operation $m(x) = (h \circ j)(x)$.

4. (13 points) Consider that $h(x) = e^{-x^2}$. Find the following operation $h^{-1}(x)$

II Limits. Read the following questions and justify your answer with a procedure.

1. (20 points) Identify the discontinuities of the function (vertical asymptotes or holes) and compute the limit using the numeric approach at those values of the function $f(x) = \frac{x^2-4}{x-2}$. Compute the limit from the left and right.

2. (20 **points**) Compute the limit using the numeric approach at those values of the function $f(x) = \frac{\sin(x)}{x}$. Compute the limit from the left and right. Use the page 4 to write your argument.

3. (10 **points**) Applying the graphical technique to compute the limit when $x \rightarrow 0$. Argument if the limit converges. Use the notion of limit from the left and right.

